

## Networking Assignment Week 1

1. Consider a case, a folder has multiple files and how would copy it to destination machine path.

- Created a folder test\_src and added files inside
- CMD USED :
  - **cp -r test\_src /home/manish/test\_dest**
- -r is used to copy all files in folder
- Verified using ls command
- Also tested copying to another user using:
  - **scp -r test\_src user2@localhost:/home/user2/**

```
manish@boss:~/Networking practice files$ pwd
/home/manish/Networking practice files
manish@boss:~/Networking practice files$ cp -r test_src /home/manish/test_dest
manish@boss:~/Networking practice files$ ls /home/manish/test_dest
f1.txt  f2.txt
```

```
manish@boss:~/Networking practice files$ mkdir test_src
manish@boss:~/Networking practice files$ cd test_src
manish@boss:~/Networking practice files$ scp -r test_src user2@localhost:/home/user2/
The authenticity of host 'localhost (:::1)' can't be established.
ED25519 key fingerprint is SHA256:ZwmmuwQEkmQJtMvWg00jxONMtseVUipc0zg5RFQ0pM.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'localhost' (ED25519) to the list of known hosts.
user2@localhost's password:
f2.txt                                100%   0    0.0KB/s   00:00
f1.txt                                100%   0    0.0KB/s   00:00
manish@boss:~/Networking practice files$ ls /home/user2/
ls: cannot open directory '/home/user2/': Permission denied
manish@boss:~/Networking practice files$ sudo ls /home/user2/
[sudo] password for manish:
Templates  test_src
manish@boss:~/Networking practice files$ sudo ls /home/user2/test_src
f1.txt  f2.txt
```

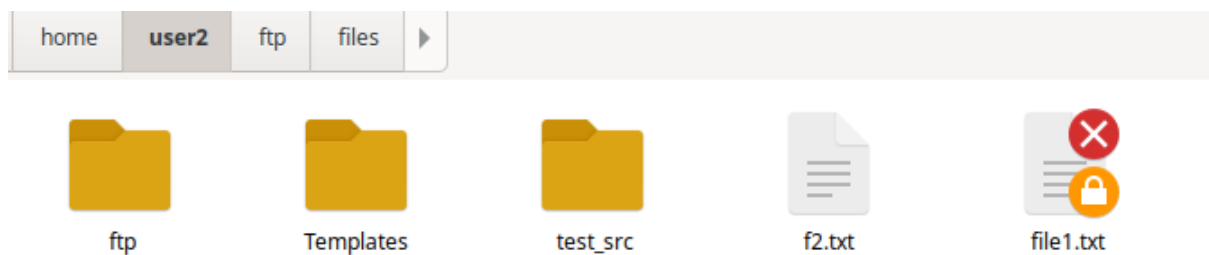
## 2. Host a FTP and SFTP server and try PUT and GET operations.

### FTP:

- Installed FTP server using vsftpd on Linux
- Started FTP service using systemctl
- Found IP address using ip a
- Connected from Windows using:
  - **ftp 192.168.1.5**
- Logged in using user2 credentials
- Uploaded file using:
  - **put file1.txt**
- Downloaded file using:
  - **get f2.txt**
- File transfer was successful (Transfer complete)

### Server (Linux):

```
manish@boss:~/Networking practice files$ sudo systemctl start vsftpd
sudo systemctl enable vsftpd
Synchronizing state of vsftpd.service with SysV service script with /lib/systemd
/systemd-sysv-install.
Executing: /lib/systemd/systemd-sysv-install enable vsftpd
manish@boss:~/Networking practice files$ sudo nano /etc/vsftpd/conf
manish@boss:~/Networking practice files$ sudo nano /etc/vsftpd.conf
manish@boss:~/Networking practice files$ sudo systemctl restart vsftpd
manish@boss:~/Networking practice files$ sudo chmod -R 777 /home/user2
manish@boss:~/Networking practice files$ ip -a
```



## Client (Windows):

```
Microsoft Windows [Version 10.0.19045.6466]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Manish>echo hello>file1.txt

C:\Users\Manish>ftp 192.168.1.5
Connected to 192.168.1.5.
220 (vsFTPd 3.0.3)
200 Always in UTF8 mode.
User (192.168.1.5:(none)): user2
331 Please specify the password.
Password:
230 Login successful.
ftp> put file1.txt
200 PORT command successful. Consider using PASV.
150 Ok to send data.
226 Transfer complete.
ftp: 7 bytes sent in 0.30Seconds 0.02Kbytes/sec.
ftp> get f2.txt
200 PORT command successful. Consider using PASV.
150 Opening BINARY mode data connection for f2.txt (9 bytes).
226 Transfer complete.
ftp: 9 bytes received in 0.00Seconds 9000.00Kbytes/sec.
ftp>
```

|                                                                                           |                  |               |      |
|-------------------------------------------------------------------------------------------|------------------|---------------|------|
|  f2    | 22-03-2026 10:05 | Text Document | 1 KB |
|  file1 | 22-03-2026 10:01 | Text Document | 1 KB |

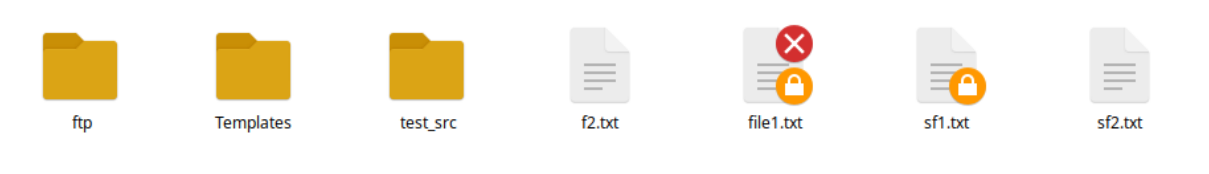
## SFTP:

- Installed SSH server using openssh-server
- Started SSH service
- Connected from Windows using:
  - **sftp user2@192.168.1.5**
- Logged in successfully
- Uploaded file using:
  - **put sf1.txt**
- Downloaded file using:
  - **get sf2.txt**
- File transfer was successful

Server (Linux):

```
manish@boss:~/Networking practice files$ sudo apt install openssh-server
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
openssh-server is already the newest version (1:9.2p1-2+deb12u6).
0 upgraded, 0 newly installed, 0 to remove and 71 not upgraded.
manish@boss:~/Networking practice files$ sudo systemctl start ssh
manish@boss:~/Networking practice files$ sudo systemctl status ssh
• ssh.service - OpenBSD Secure Shell server
  Loaded: loaded (/lib/systemd/system/ssh.service; enabled; preset: enabled)
  Active: active (running) since Mon 2026-03-23 09:53:07 +07; 19min ago
    Docs: man:sshd(8)
          man:sshd_config(5)
  Main PID: 851 (sshd)
    Tasks: 1 (limit: 8388)
  Memory: 4.9M
    CPU: 30ms
  CGroup: /system.slice/ssh.service
          └─851 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Mar 23 09:53:07 boss systemd[1]: Starting ssh.service - OpenBSD Secure Shell ser
Mar 23 09:53:07 boss sshd[851]: Server listening on 0.0.0.0 port 22.
Mar 23 09:53:07 boss sshd[851]: Server listening on :: port 22.
Mar 23 09:53:07 boss systemd[1]: Started ssh.service - OpenBSD Secure Shell ser
...skipping...
• ssh.service - OpenBSD Secure Shell server
  Loaded: loaded (/lib/systemd/system/ssh.service; enabled; preset: enabled)
  Active: active (running) since Mon 2026-03-23 09:53:07 +07; 19min ago
```



Client (Windows):

```
Microsoft Windows [Version 10.0.19045.6466]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Manish>sftp user2@192.168.1.5
user2@192.168.1.5's password:
Connected to 192.168.1.5.
sftp> put sf1.txt
Uploading sf1.txt to /home/user2/sf1.txt
sf1.txt                                     100%   5    0.0KB/s   00:00
sftp> get sf2.txt
Fetching /home/user2/sf2.txt to sf2.txt
sf2.txt                                     100%   6    0.4KB/s   00:00
sftp>
```

|     |                  |               |      |
|-----|------------------|---------------|------|
| sf1 | 22-03-2026 10:11 | Text Document | 1 KB |
| sf2 | 22-03-2026 10:14 | Text Document | 1 KB |

### 3. Explore with Wireshark/TCP-dump/cisco packet tracer tools and learn about packet filters.

- Opened Wireshark and selected network interface
- Captured live network traffic
- Generated traffic by browsing websites
- Applied packet filters to analyze traffic

#### Filters used:

- tcp
- ip.addr == 192.168.1.5
- tcp.port == 443

#### Tcp:

| No. | Time        | Source                | Destination           | Protocol | Length | Info                                                                                                  |
|-----|-------------|-----------------------|-----------------------|----------|--------|-------------------------------------------------------------------------------------------------------|
| 18  | 0.40955774  | 2401:4900:1ce0:1424.. | 2404:6800:4007:82e... | TCP      | 86     | 32884 → 443 [ACK] Seq=4286 Ack=1500 Win=541 Len=0 TSval=2983388078 TSecr=1199332427                   |
| 21  | 1.791170408 | 2401:4900:1ce0:1424.. | 2606:4700:90d3:c0c9.. | TCP      | 86     | 58102 → 443 [ACK] Seq=1 Ack=1 Win=631 Len=0 TSval=1790890515 TSecr=3445902244                         |
| 22  | 3.07789155  | 2606:4700:90d3:c0c9.. | 2401:4900:1ce0:1424.. | TCP      | 86     | [TCP ACKed unseen segment] 443 → 58102 (ACK) Seq=1 Ack=2 Win=17 Len=0 TSval=3445940735 TSecr=17007990 |
| 23  | 3.839174340 | 2401:4900:1ce0:1424.. | 2404:4e42:200:649..   | TCP      | 86     | 45374 → 443 [ACK] Seq=1 Ack=1 Win=641 Len=0 TSval=1321443748 TSecr=329898962                          |
| 24  | 3.839174340 | 2401:4900:1ce0:1424.. | 2404:4e42:200:649..   | TCP      | 86     | [TCP ACKed unseen segment] 443 → 45374 (ACK) Seq=1 Ack=2 Win=271 Len=0 TSval=329944014 TSecr=13210355 |
| 25  | 5.059813901 | 35.82.117.226         | 192.168.1.5           | TLsv1.2  | 105    | Application Data                                                                                      |
| 26  | 5.059825555 | 35.82.117.226         | 192.168.1.5           | TLsv1.2  | 90     | Application Data                                                                                      |
| 27  | 5.059825705 | 35.82.117.226         | 192.168.1.5           | TCP      | 66     | 443 → 42070 [FIN, ACK] Seq=64 Ack=1 Win=197 Len=0 TSval=2745191385 TSecr=797838485                    |
| 28  | 5.059973111 | 192.168.1.5           | 35.82.117.226         | TCP      | 66     | 42070 → 443 [ACK] Seq=1 Ack=65 Win=595 Len=0 TSval=797847481 TSecr=2745191385                         |
| 29  | 5.059920495 | 192.168.1.5           | 35.82.117.226         | TCP      | 66     | 42070 → 443 [FIN, ACK] Seq=1 Ack=65 Win=595 Len=0 TSval=797847482 TSecr=2745191385                    |
| 30  | 6.052363200 | 35.82.117.226         | 192.168.1.5           | TCP      | 66     | 443 → 42070 [ACK] Seq=65 Ack=2 Win=197 Len=0 TSval=2745191654 TSecr=797847482                         |
| 31  | 6.052374662 | 35.82.117.226         | 192.168.1.5           | TLsv1.2  | 105    | Application Data                                                                                      |
| 32  | 6.052374842 | 35.82.117.226         | 192.168.1.5           | TLsv1.2  | 90     | Application Data                                                                                      |
| 33  | 6.052375012 | 35.82.117.226         | 192.168.1.5           | TCP      | 66     | 443 → 42082 [FIN, ACK] Seq=64 Ack=1 Win=121 Len=0 TSval=2745191682 TSecr=797838537                    |
| 34  | 6.095170521 | 192.168.1.5           | 35.82.117.226         | TCP      | 66     | 42082 → 443 [ACK] Seq=1 Ack=65 Win=574 Len=0 TSval=797848517 TSecr=2745191682                         |
| 35  | 6.266393370 | 192.168.1.5           | 9.9.9.9               | TCP      | 74     | 37364 → 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1400 SACK_PERM TSval=202415827 TSecr=0 WS=128             |
| 36  | 7.071542680 | 9.9.9.9               | 192.168.1.5           | TCP      | 74     | 443 → 37364 [SYN, ACK] Seq=0 Ack=1 Win=43440 Len=0 MSS=1400 SACK_PERM TSval=2294314462 TSecr=20241582 |
| 37  | 7.071624924 | 192.168.1.5           | 9.9.9.9               | TCP      | 66     | 37364 → 443 [ACK] Seq=1 Ack=1 Win=64256 Len=0 TSval=202416632 TSecr=2294314462                        |

#### ip

| No. | Time        | Source        | Destination   | Protocol | Length | Info                                                                                                  |
|-----|-------------|---------------|---------------|----------|--------|-------------------------------------------------------------------------------------------------------|
| 25  | 5.059813901 | 35.82.117.226 | 192.168.1.5   | TLsv1.2  | 105    | Application Data                                                                                      |
| 26  | 5.059825555 | 35.82.117.226 | 192.168.1.5   | TLsv1.2  | 90     | Application Data                                                                                      |
| 27  | 5.059825705 | 35.82.117.226 | 192.168.1.5   | TCP      | 66     | 443 → 42070 [FIN, ACK] Seq=64 Ack=1 Win=197 Len=0 TSval=2745191385 TSecr=797838485                    |
| 28  | 5.059973111 | 192.168.1.5   | 35.82.117.226 | TCP      | 66     | 42070 → 443 [ACK] Seq=1 Ack=65 Win=595 Len=0 TSval=797847481 TSecr=2745191385                         |
| 29  | 5.059920495 | 192.168.1.5   | 35.82.117.226 | TCP      | 66     | 42070 → 443 [FIN, ACK] Seq=1 Ack=65 Win=595 Len=0 TSval=797847482 TSecr=2745191385                    |
| 30  | 6.052363200 | 35.82.117.226 | 192.168.1.5   | TCP      | 66     | 443 → 42070 [ACK] Seq=65 Ack=2 Win=197 Len=0 TSval=2745191654 TSecr=797847482                         |
| 31  | 6.052374662 | 35.82.117.226 | 192.168.1.5   | TLsv1.2  | 105    | Application Data                                                                                      |
| 32  | 6.052374842 | 35.82.117.226 | 192.168.1.5   | TLsv1.2  | 90     | Application Data                                                                                      |
| 33  | 6.052375012 | 35.82.117.226 | 192.168.1.5   | TCP      | 66     | 443 → 42082 [FIN, ACK] Seq=64 Ack=1 Win=121 Len=0 TSval=2745191682 TSecr=797838537                    |
| 34  | 6.095170521 | 192.168.1.5   | 35.82.117.226 | TCP      | 66     | 42082 → 443 [ACK] Seq=1 Ack=65 Win=574 Len=0 TSval=797848517 TSecr=2745191682                         |
| 35  | 6.266393370 | 192.168.1.5   | 9.9.9.9       | TCP      | 74     | 37364 → 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1400 SACK_PERM TSval=202415827 TSecr=0 WS=128             |
| 36  | 7.071542680 | 9.9.9.9       | 192.168.1.5   | TCP      | 74     | 443 → 37364 [SYN, ACK] Seq=0 Ack=1 Win=43440 Len=0 MSS=1400 SACK_PERM TSval=2294314462 TSecr=20241582 |
| 37  | 7.071624924 | 192.168.1.5   | 9.9.9.9       | TCP      | 66     | 37364 → 443 [ACK] Seq=1 Ack=1 Win=64256 Len=0 TSval=202416632 TSecr=2294314462                        |
| 37  | 7.081287920 | 192.168.1.5   | 9.9.9.9       | TLsv1.3  | 583    | Client Hello                                                                                          |
| 37  | 7.081287920 | 192.168.1.5   | 192.168.1.1   | DNS      | 70     | Standard query 0xf74c A dns.google                                                                    |
| 39  | 7.088483357 | 192.168.1.5   | 192.168.1.1   | DNS      | 70     | Standard query response 0xb2b3 AAAA dns.google                                                        |
| 40  | 7.088503825 | 192.168.1.5   | 192.168.1.1   | DNS      | 102    | Standard query response 0xf74c A dns.google A 8.8.4.4 A 8.8.8                                         |
| 41  | 7.101686228 | 192.168.1.1   | 192.168.1.5   | DNS      | 126    | Standard query response 0xb2b3 AAAA dns.google AAAA 2001:4860:4860::8888 AAAA 2001:4860:4860::8844    |
| 42  | 7.101697149 | 192.168.1.1   | 192.168.1.5   | DNS      | 66     | 443 → 37364 [ACK] Seq=1 Ack=518 Win=43008 Len=0 TSval=2294315187 TSecr=202416642                      |
| 46  | 7.168591041 | 9.9.9.9       | 192.168.1.5   | TCP      | 66     | 443 → 37364 [ACK] Seq=1 Ack=518 Win=43008 Len=0 TSval=2294315187 TSecr=202416642                      |

#### port

| No. | Time        | Source                | Destination           | Protocol | Length | Info                                                                                                     |
|-----|-------------|-----------------------|-----------------------|----------|--------|----------------------------------------------------------------------------------------------------------|
| 8   | 0.004827505 | 2401:4900:1ce0:1424.. | 2404:6800:4007:82e... | TCP      | 1454   | 32884 → 443 [ACK] Seq=2811 Ack=326 Win=541 Len=1368 TSval=2983387673 TSecr=1199331967 [TCP segment of... |
| 9   | 0.004828677 | 2401:4900:1ce0:1424.. | 2404:6800:4007:82e... | TLsv1.2  | 193    | Application Data                                                                                         |
| 10  | 0.019047079 | 2404:6800:4007:82e... | 2401:4900:1ce0:1424.. | TCP      | 98     | [TCP Dup ACK #1] 443 → 32884 (ACK) Seq=326 Ack=1 Win=1096 Len=0 TSval=1199332073 TSecr=2983370693 SL...  |
| 11  | 0.019058280 | 2404:6800:4007:82e... | 2401:4900:1ce0:1424.. | TCP      | 86     | 443 → 32884 [ACK] Seq=326 Ack=75 Win=1096 Len=0 TSval=1199332073 TSecr=2983387669                        |
| 12  | 0.021168399 | 2404:6800:4007:82e... | 2401:4900:1ce0:1424.. | TCP      | 86     | 443 → 32804 [ACK] Seq=326 Ack=1449 Win=1091 Len=0 TSval=1199332075 TSecr=2983387673                      |
| 13  | 0.021179430 | 2404:6800:4007:82e... | 2401:4900:1ce0:1424.. | TCP      | 86     | 443 → 32884 [ACK] Seq=326 Ack=2811 Win=1086 Len=0 TSval=1199332075 TSecr=2983387673                      |
| 14  | 0.022621970 | 2404:6800:4007:82e... | 2401:4900:1ce0:1424.. | TCP      | 86     | 443 → 32884 [ACK] Seq=326 Ack=4179 Win=1085 Len=0 TSval=1199332077 TSecr=2983387673                      |
| 15  | 0.025229821 | 2404:6800:4007:82e... | 2401:4900:1ce0:1424.. | TCP      | 86     | 443 → 32884 [ACK] Seq=326 Ack=4286 Win=1085 Len=0 TSval=1199332079 TSecr=2983387673                      |
| 16  | 0.499773453 | 2404:6800:4007:82e... | 2401:4900:1ce0:1424.. | TLsv1.2  | 1043   | Application Data                                                                                         |
| 17  | 0.409700126 | 2404:6800:4007:82e... | 2401:4900:1ce0:1424.. | TLsv1.2  | 303    | Application Data                                                                                         |
| 18  | 0.40955774  | 2401:4900:1ce0:1424.. | 2404:6800:4007:82e... | TCP      | 86     | 32884 → 443 [ACK] Seq=4286 Ack=1500 Win=541 Len=0 TSval=2983388078 TSecr=1199332427                      |
| 21  | 1.791170408 | 2401:4900:1ce0:1424.. | 2606:4700:90d3:c0c9.. | TCP      | 86     | 58102 → 443 [ACK] Seq=1 Ack=1 Win=631 Len=0 TSval=1790890515 TSecr=3445902244                            |
| 22  | 3.07789155  | 2606:4700:90d3:c0c9.. | 2401:4900:1ce0:1424.. | TCP      | 86     | [TCP ACKed unseen segment] 443 → 58102 (ACK) Seq=1 Ack=2 Win=17 Len=0 TSval=3445940735 TSecr=17007990    |
| 23  | 3.839174340 | 2401:4900:1ce0:1424.. | 2404:4e42:200:649..   | TCP      | 86     | 45374 → 443 [ACK] Seq=1 Ack=1 Win=641 Len=0 TSval=1321443748 TSecr=329898962                             |
| 24  | 3.839174340 | 2401:4900:1ce0:1424.. | 2404:4e42:200:649..   | TCP      | 86     | [TCP ACKed unseen segment] 443 → 45374 (ACK) Seq=1 Ack=2 Win=271 Len=0 TSval=329944014 TSecr=13210355    |
| 25  | 5.059813901 | 35.82.117.226         | 192.168.1.5           | TLsv1.2  | 105    | Application Data                                                                                         |
| 26  | 5.059825555 | 35.82.117.226         | 192.168.1.5           | TLsv1.2  | 90     | Application Data                                                                                         |
| 27  | 5.059825705 | 35.82.117.226         | 192.168.1.5           | TCP      | 66     | 443 → 42070 [FIN, ACK] Seq=64 Ack=1 Win=197 Len=0 TSval=2745191385 TSecr=797838485                       |
| 28  | 5.059973111 | 192.168.1.5           | 35.82.117.226         | TCP      | 66     | 42070 → 443 [ACK] Seq=1 Ack=65 Win=595 Len=0 TSval=797847481 TSecr=2745191385                            |

## ii) Using tcp-dump command:

```
manish@boss:~/Networking practice files$ sudo tcpdump -i wlo1
[sudo] password for manish:
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on wlo1, link-type EN10MB (Ethernet), snapshot length 262144 bytes
11:19:55.616380 IP6 boss.44788 > 2606:4700:8392:7cbc:c2a5:0:5ff2:75c4.https: Flags [.], seq 3871726348:3871727696, ack 2535538181, win 3093, options [nop,nop,TS val 1885112234 ecr 1972142801], length 1348
11:19:55.616399 IP6 boss.44788 > 2606:4700:8392:7cbc:c2a5:0:5ff2:75c4.https: Flags [P.], seq 1348:2696, ack 1, win 3093, options [nop,nop,TS val 1885112234 ecr 1972142801], length 1348
11:19:55.616405 IP6 boss.44788 > 2606:4700:8392:7cbc:c2a5:0:5ff2:75c4.https: Flags [.], seq 2696:4044, ack 1, win 3093, options [nop,nop,TS val 1885112234 ecr 1972142801], length 1348
11:19:55.616407 IP6 boss.44788 > 2606:4700:8392:7cbc:c2a5:0:5ff2:75c4.https: Flags [P.], seq 4044:5392, ack 1, win 3093, options [nop,nop,TS val 1885112234 ecr 1972142801], length 1348
```

## Protocols:

```
manish@boss:~/Networking practice files$ sudo tcpdump -i wlo1 tcp
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on wlo1, link-type EN10MB (Ethernet), snapshot length 262144 bytes
11:20:51.133991 IP6 boss.48174 > 2001:4860:482d:7700::.https: Flags [.], ack 2982411237, win 528, options [nop,nop,TS val 12208788585 ecr 1772304348], length 0
11:20:51.241372 IP6 2001:4860:482d:7700::.https > boss.48174: Flags [R], seq 2982411237, win 0, length 0
11:20:57.352401 IP6 2a06:98c1:3107::6812:2715.https > boss.36758: Flags [P.], seq 3110166946:3110166970, ack 531637127, win 17, options [nop,nop,TS val 2814875182 ecr 2529127772], length 24
11:20:57.352455 IP6 boss.36758 > 2a06:98c1:3107::6812:2715.https: Flags [.], ack 24, win 481, options [nop,nop,TS val 2529141757 ecr 2814875182], length 0
11:20:57.352772 IP6 boss.36758 > 2a06:98c1:3107::6812:2715.https: Flags [P.], seq 1:29, ack 24, win 481, options [nop,nop,TS val 2529141758 ecr 2814875182], length 28
11:20:57.693779 IP6 boss.36758 > 2a06:98c1:3107::6812:2715.https: Flags [P.], seq 1:29, ack 24, win 481, options [nop,nop,TS val 2529142099 ecr 2814875182], length 28
```

## Ip:

```
manish@boss:~/Networking practice files$ sudo tcpdump -i wlo1 host 192.168.1.5
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on wlo1, link-type EN10MB (Ethernet), snapshot length 262144 bytes
11:21:49.428452 IP _gateway.bootps > boss.bootpc: BOOTP/DHCP, Reply, length 548
11:21:49.441955 ARP, Request who-has _gateway tell boss, length 28
11:21:49.443956 ARP, Reply _gateway is-at 30:4f:75:c5:bc:cf (oui Unknown), length 28
11:21:49.443979 IP boss.52353 > _gateway.domain: 14629+ A? dns.google. (28)
11:21:49.444000 IP boss.52353 > _gateway.domain: 58156+ AAAA? dns.google. (28)
11:21:49.444005 IP boss > _gateway: ICMP boss udp port bootpc unreachable, length 556
11:21:49.459972 IP _gateway.domain > boss.52353: 14629 2/0/0 A 8.8.4.4, A 8.8.8.8 (60)
11:21:49.463746 IP _gateway.domain > boss.52353: 58156 2/0/0 AAAA 2001:4860:4860::8888, AAAA 2001:4860:4860::8844 (84)
11:21:49.518993 IP boss.50938 > _gateway.domain: 65527+ PTR? 5.1.168.192.in-addr.arpa. (42)
11:21:49.521812 IP _gateway.domain > boss.50938: 65527 NXDomain 0/0/0 (42)
11:21:49.522383 IP boss.48122 > _gateway.domain: 65363+ PTR? 1.1.168.192.in-addr.arpa. (42)
```

## Port:

```
manish@boss:~/Networking practice files$ sudo tcpdump -i wlo1 port 443
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on wlo1, link-type EN10MB (Ethernet), snapshot length 262144 bytes
11:22:31.521996 IP6 boss.41138 > maa03s21-in-x0a.1e100.net.https: Flags [.], ack 2617067187, win 659, options [nop,nop,TS val 498885951 ecr 3634254510], length 0
11:22:31.650263 IP6 maa03s21-in-x0a.1e100.net.https > boss.41138: Flags [.], ack 1, win 1045, options [nop,nop,TS val 3634299622 ecr 498658118], length 0
11:22:32.202700 IP6 boss.36566 > lcmaaa-az-in-x0e.1e100.net.https: Flags [P.], seq 4282374861:4282374896, ack 2518512247, win 3552, options [nop,nop,TS val 326836148 ecr 1152962207], length 35
11:22:32.205874 IP6 boss.35002 > maa05s13-in-x03.1e100.net.https: UDP, length 99
11:22:32.206444 IP6 boss.35002 > maa05s13-in-x03.1e100.net.https: UDP, length 99
```

#### 4. Understand linux utility commands like - ping, arp (Understand each params from ifconfig output)

##### ping:

- Used to check network connectivity
- Command used:

ping [google.com](https://www.google.com)

- Observed replies from server with time delay
- Shows packets sent, received and packet loss

```
manish@boss:~/Networking practice files$ ping google.com
PING google.com(pnmetaa-bb-in-x0e.1e100.net (2404:6800:4007:805::200e)) 56 data bytes
64 bytes from pnmetaa-bb-in-x0e.1e100.net (2404:6800:4007:805::200e): icmp_seq=1 ttl=118 time=861 ms
64 bytes from pnmetaa-bb-in-x0e.1e100.net (2404:6800:4007:805::200e): icmp_seq=3 ttl=118 time=870 ms
64 bytes from pnmetaa-bb-in-x0e.1e100.net (2404:6800:4007:805::200e): icmp_seq=5 ttl=118 time=96.5 ms
64 bytes from pnmetaa-bb-in-x0e.1e100.net (2404:6800:4007:805::200e): icmp_seq=6 ttl=118 time=505 ms
^C
--- google.com ping statistics ---
8 packets transmitted, 4 received, 50% packet loss, time 11215ms
rtt min/avg/max/mdev = 96.546/583.159/870.267/317.201 ms
```

##### arp:

- Used to find MAC address of devices in network
- Command used:

arp -a

- Displays IP address to MAC address mapping
- Example: gateway IP mapped to MAC address

```
manish@boss:~/Networking practice files$ sudo arp
Address          HWtype  HWaddress      Flags Mask    Iface
_gateway         ether    30:4f:75:c5:bc:cf  C             wlo1
manish@boss:~/Networking practice files$ sudo arp -a
_gateway (192.168.1.1) at 30:4f:75:c5:bc:cf [ether] on wlo1
```

## ifconfig:

- Used to view network interface details
- Command used:

ifconfig

### Parameters observed:

- **inet** → IP address (192.168.1.5)
- **ether** → MAC address
- **mtu** → Maximum Transmission Unit (1500)
- **RX/TX** → packets received and transmitted

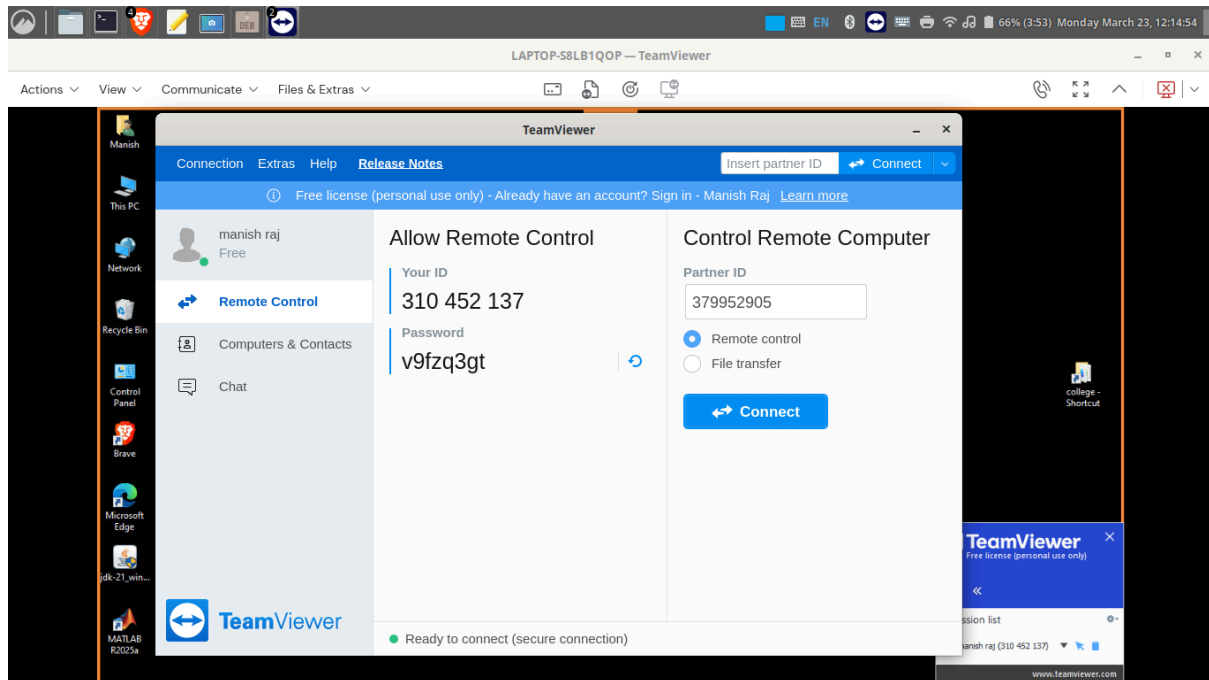
```
manish@boss:~/Networking practice files$ sudo ifconfig
lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 152 bytes 10673 (10.4 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 152 bytes 10673 (10.4 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

wlo1: flags=4099<UP,BROADCAST,MULTICAST> mtu 1500
    inet 192.168.1.5 netmask 255.255.255.0 broadcast 192.168.1.255
    inet6 2401:4900:1ce0:1424:ee49:a47b:1e6a:a9e7 prefixlen 64 scopeid 0x0<global>
    inet6 fe80::3745:56f:2c7f:63fd prefixlen 64 scopeid 0x20<link>
    ether 20:58:43:e6:8e:bc txqueuelen 1000 (Ethernet)
    RX packets 332601 bytes 456713429 (435.5 MiB)
    RX errors 0 dropped 557 overruns 0 frame 0
    TX packets 54412 bytes 14748456 (14.0 MiB)
    TX errors 0 dropped 61 overruns 0 carrier 0 collisions 0
```

## 5. Understand what happens when duplicate IPs are configured in a network.

- When duplicate IP addresses are configured, two or more devices share the same IP
- This causes network confusion and communication issues
  - Packet loss occurs
  - Network becomes unstable
  - ARP conflicts happen (wrong MAC mapping)
  - Devices may not connect to network properly

## 6. Understand how to access remote system using (VNC viewer, Anydesk, teamviewer and remote desktop connections)



- Installed TeamViewer on both systems
- Opened application and obtained ID and password
- Entered remote ID in client system
- Established connection successfully
- Accessed and controlled remote system

## 7. How to check if your default gateway is reachable or not and understand the default gateway.

- Default gateway is the router used to connect the local network to other networks (internet)
- Found default gateway using:

ip route

- Identified gateway IP as 192.168.1.1
- Checked reachability using:

ping 192.168.1.1

- Received replies with 0% packet loss, confirming the gateway is reachable

```

manish@boss:~$ ip route
default via 192.168.1.1 dev wlo1 proto dhcp src 192.168.1.5 metric 600
192.168.1.0/24 dev wlo1 proto kernel scope link src 192.168.1.5 metric 600
manish@boss:~$ ping 192.168.1.1
PING 192.168.1.1 (192.168.1.1) 56(84) bytes of data.
64 bytes from 192.168.1.1: icmp_seq=1 ttl=64 time=10.6 ms
64 bytes from 192.168.1.1: icmp_seq=2 ttl=64 time=10.6 ms
64 bytes from 192.168.1.1: icmp_seq=3 ttl=64 time=1.58 ms
64 bytes from 192.168.1.1: icmp_seq=4 ttl=64 time=6.36 ms
64 bytes from 192.168.1.1: icmp_seq=5 ttl=64 time=2.39 ms
^C
--- 192.168.1.1 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4006ms
rtt min/avg/max/mdev = 1.579/6.313/10.636/3.870 ms
manish@boss:~$ █

```

8. Check `iwconfig`/`ifconfig` to understand in detail about network interfaces (check about interface speed, MTU and other parameters)

- Checked network interface details using:

```

ifconfig
iwconfig

```

From `ifconfig`:

- `inet` → IP address (192.168.1.5)
- `ether` → MAC address (20:58:43:e6:8e:bc)
- `mtu` → Maximum Transmission Unit (1500)
- `RX/TX` → packets received and transmitted

From `iwconfig`:

- `ESSID` → Connected WiFi network (Airtel\_siva\_3012)
- `Bit Rate` → Interface speed (292.6 Mb/s)
- `Frequency` → 5.22 GHz
- `Link Quality` → signal strength

```
manish@boss:~$ sudo ifconfig
lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    loop txqueuelen 1000 (Local Loopback)
    RX packets 11536 bytes 8877615 (8.4 MiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 11536 bytes 8877615 (8.4 MiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

wlo1: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.1.5 netmask 255.255.255.0 broadcast 192.168.1.255
    ether 20:58:43:e6:8e:bc txqueuelen 1000 (Ethernet)
    RX packets 546222 bytes 727563970 (693.8 MiB)
    RX errors 0 dropped 1219 overruns 0 frame 0
    TX packets 112380 bytes 34529135 (32.9 MiB)
    TX errors 0 dropped 84 overruns 0 carrier 0 collisions 0
```

```
manish@boss:~$ sudo iwconfig
lo          no wireless extensions.

wlo1       IEEE 802.11  ESSID:"Airtel_siva_3012"
Mode:Managed  Frequency:5.22 GHz  Access Point: 30:4F:75:C5:BC:D1
Bit Rate=292.6 Mb/s   Tx-Power=30 dBm
Retry short limit:7   RTS thr:off   Fragment thr:off
Encryption key:off
Power Management:on
Link Quality=46/70  Signal level=-64 dBm
Rx invalid nwid:0  Rx invalid crypt:0  Rx invalid frag:0
Tx excessive retries:1 Invalid misc:15  Missed beacon:0
```

9. Log in to your home router's web interface (usually at 192.168.1.1 or 192.168.0.1) and check the connected devices list.

- Opened web browser and entered router IP address (192.168.1.1)
- Logged in using router credentials
- Navigated to “Current Wireless Users” section
- Viewed list of connected devices
- Observed details like IP address, MAC address and device name

Not secure https://192.168.1.1/status/current-wireless-user

Search For Menu

Status / Current Wireless Users

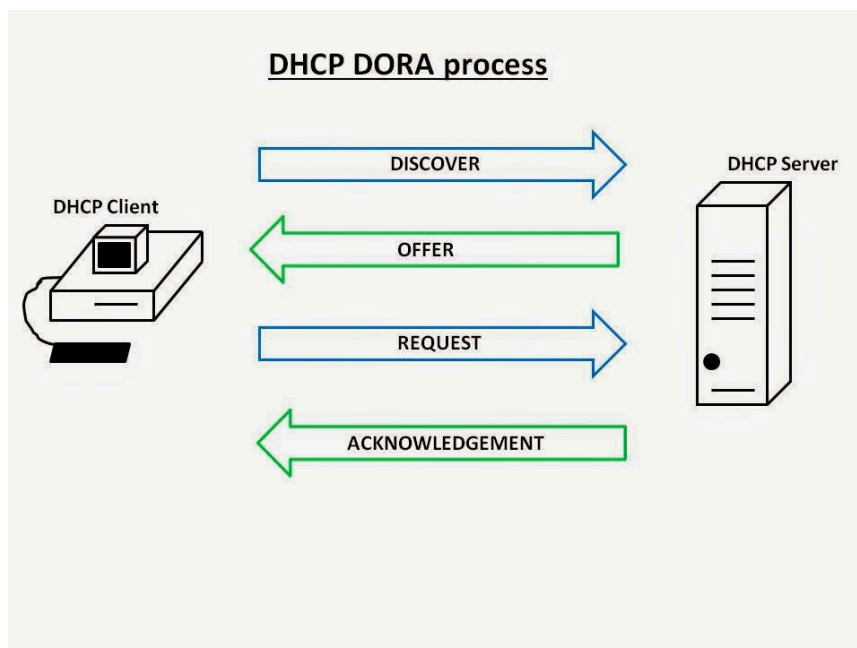
Current Wireless Users

WiFi Band: 2.4G SSID Index: 1

Total Clients: 1

| # | MAC               | IP          | Host Name       | Receiving Rate (Mbit/s) | Sending Rate (Mbit/s) | Duration (s) | RSSI (dBm) |
|---|-------------------|-------------|-----------------|-------------------------|-----------------------|--------------|------------|
| 1 | 40:9F:38:63:D4:F9 | 192.168.1.4 | LAPTOP-S8LB1QOP | 150                     | 17                    | 9414         | -67        |

10. Explain how a DHCP server assigns IP addresses to devices in your network.



- DHCP (Dynamic Host Configuration Protocol) automatically assigns IP addresses to devices in a network

Working process (DORA):

- Device sends DHCP Discover message to find server
- DHCP server sends DHCP Offer with available IP address
- Device sends DHCP Request to accept the IP
- Server sends DHCP Acknowledgement (ACK) confirming assignment
- The device is then assigned an IP address and can communicate in the network

## 11. Using a terminal, connect to a remote machine via SSH and telnet.

### SSH:

- Installed and started SSH service on Linux
- Connected from Windows using:

`ssh user2@192.168.1.5`

- Logged in using credentials
- Successfully accessed remote terminal

```
manish@boss:~$ sudo systemctl start ssh
```

```
Microsoft Windows [Version 10.0.19045.6466]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Manish>ssh user2@192.168.1.5
user2@192.168.1.5's password:
Linux boss 6.12.43+deb12-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.12.43-1~bpo12+1 (2025-09-06) x86_64

The programs included with the BOSS GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

BOSS GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
user2@boss:~$ pwd
/home/user2
user2@boss:~$ ls
f2.txt  file1.txt  file2.txt  templates  test_ssh
user2@boss:~$
```

### Telnet:

- Installed Telnet server using openbsd-inetd
- Enabled Telnet service and started inetd
- Verified port 23 is listening
- Enabled Telnet client on Windows
- Connected using:

`telnet 192.168.1.5`

- Successfully accessed remote system

```
manish@boss:~$ sudo systemctl start openbsd-inetd
```

```
manish@boss:~$ sudo netstat -tulnp | grep :23
tcp        0      0 0.0.0.0:23          0.0.0.0:*          LISTEN     14154/inetd
tcp6       0      0 :::23             :::*                LISTEN     14154/inetd
```

```
C:\WINDOWS\system32>dism /online /Enable-Feature /FeatureName:TelnetClient
```

```
Deployment Image Servicing and Management tool
```

```
Version: 10.0.19041.3636
```

```
Image Version: 10.0.19045.6466
```

```
Enabling feature(s)
```

```
[=====100.0%=====]
```

```
The operation completed successfully.
```

```
C:\WINDOWS\system32>telnet 192.168.1.5
```